FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 — SUB-STRAND 3.3: ANIMAL GASEOUS EXCHANGE AND RESPIRATION	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION ASSESSMENT — "Why Does My Body Work So Much Harder to Get Oxygen Into My Cells During Exercise — And What Happens Inside Those Cells When There Is Not Enough?"

HOW TO USE THIS DOCUMENT

This document is your guide for writing a high-quality Final Explanation. A Final Explanation is your chance to show everything you have figured out across this unit — using evidence, scientific vocabulary, and logical reasoning to fully answer the driving question. Read each section prompt carefully, study the exemplar response beneath it, then write your own answer in your exercise book or on the answer sheet provided by your teacher.

YOUR GOAL: Explain the complete journey of oxygen — from the air outside your body, across your respiratory surface, through your blood, and into your muscle cells — and explain what happens when that oxygen supply runs out during intense exercise like the step-ups you performed on Day 1.

BEFORE YOU WRITE — REVIEW YOUR EVIDENCE:

– Your resting and post-exercise breathing rate and pulse rate data from the class data table
– Your diagrams of the human respiratory system and alveoli
– Your notes and results from the breathing mechanics investigation (chest volume and pressure)
– Your balanced equations for aerobic and anaerobic respiration
– Your Driving Question Board (DQB) — check which questions have been answered and how

SCIENTIFIC VOCABULARY YOU SHOULD USE:

respiratory surface, alveoli, diffusion, concentration gradient, tidal volume, ventilation rate, haemoglobin, circulatory system, aerobic respiration, ATP, glucose, anaerobic respiration, lactic acid, oxygen debt, diaphragm, intercostal muscles, partial pressure

KICD LEARNING OUTCOMES ADDRESSED:

By the end of this unit, you should be able to:

1. Explain how animals exchange gases with their environment

2. Explain how animals obtain energy from food 3. Relate the structure of the human respiratory system to its function 4. Distinguish between aerobic and anaerobic respiration and their products

Section 1: The Phenomenon — What Your Body Did During the Step-Ups	
Go back to the very first day of this unit. You measured your breathing rate and pulse rate at rest, then again immediately after 2 minutes of vigorous step-ups. Describe what you personally observed. What changed, and why was it puzzling? Use your class data to support your answer. What questions did these changes raise for you that you could not yet answer at the time?	At rest, my breathing rate was about 14 breaths per minute and my pulse was 72 beats per minute. After 2 minutes of step-ups, my breathing rate jumped to 28 breaths per minute and my pulse rose to over 120 beats per minute — both roughly doubled. This was puzzling because even though I was breathing hard the whole time, my leg muscles still felt a painful burning sensation. I could not explain why my heart was speeding up at the same time as my lungs, or where exactly the oxygen was going once it entered my chest. I also could not explain why my muscles burned if I had been breathing throughout. These observations raised questions I could not yet answer: What is the link between the lungs and the heart? Where does oxygen actually go after entering the lungs? Why do muscles burn during intense exercise? These became the questions our class tracked on the Driving Question Board throughout the unit.

Section 2: The Journey of Oxygen — From Air to Alveolus to Blood	
Explain how oxygen gets from the air outside your body into your bloodstream. In your answer, describe: (a) the key features of the human respiratory system that move air inward, (b) the structural features of the alveoli that make them an efficient respiratory surface, and (c) the process by which oxygen crosses into the blood. Use scientific vocabulary and refer to at least ONE structural feature and ONE process by name.	When you breathed in during the step-ups, your diaphragm contracted and flattened downward while your external intercostal muscles pulled your ribcage upward and outward. This increased the volume of your thoracic cavity, which decreased the air pressure inside your lungs below atmospheric pressure, so air rushed in — this is called inspiration. Air travelled down the trachea, through the bronchi and bronchioles, and reached the alveoli — tiny air sacs deep in the lungs. The alveoli are an excellent respiratory surface because they have an enormous combined surface area (estimated at around 70 m^2 in an adult — roughly the size of a full basketball court), walls only one cell thick, a rich network of capillaries surrounding them, and a moist lining that allows gases to dissolve. Because oxygen concentration is high in the alveolar air and low in the blood arriving from the body, there is a steep concentration gradient. Oxygen diffuses — moves passively from a region of high concentration to low concentration — across the thin alveolar and capillary walls and into the blood. There, it binds to haemoglobin in red blood cells, forming oxyhaemoglobin, and is carried to working muscles. During exercise, you breathed faster and more deeply (increased ventilation rate and tidal volume), which kept the concentration gradient steep so diffusion could continue at a high rate.

Section 3: The Role of the Circulatory System — Why Your Heart Also Sped Up	
Many learners were puzzled on Day 1 about why their pulse	The respiratory system and the circulatory system work as a team. The lungs load oxygen into the blood, but it is the heart and blood vessels that deliver that oxygen to cells. During the step-ups, your leg muscle cells were breaking down glucose rapidly to

rate rose at the same time as their breathing rate. Explain the connection between the respiratory system and the circulatory system during exercise. Why does the heart need to beat faster when muscles are working hard? What does the blood carry, and why does delivery speed matter?	release energy (ATP). This process — aerobic respiration — consumed oxygen and produced carbon dioxide as a waste product. As oxygen levels in the muscle capillaries dropped and carbon dioxide levels rose, your brain detected these chemical changes and sent signals to speed up your heart rate. A faster heart beat pumped oxygenated blood to your muscles more quickly, and also removed carbon dioxide more quickly back to the lungs to be exhaled. So the rise in pulse rate you measured — from around 72 bpm to over 120 bpm — was your circulatory system increasing the delivery rate of oxygen and the removal rate of carbon dioxide to keep pace with the energy demands of the exercise. This is why both rates rose together: the lungs and the heart are linked in one continuous loop. Think of it like the delivery trucks (blood) having to make more trips per minute because the factory (your muscles) suddenly needs far more raw materials (oxygen and glucose).

Section 4: Inside the Muscle Cell — Aerobic and Anaerobic Respiration	
Explain what happens inside your muscle cells during exercise. Your answer must: (a) describe aerobic respiration, including the reactants, products, and purpose of the reaction; (b) explain why the burning sensation in your legs occurs, using the term anaerobic respiration; and (c) give the word equations (or balanced chemical equations if you can) for both processes. Explain what 'oxygen debt' means and how your body repays it.	Inside your muscle cells, glucose from digested food (like ugali or githeri) is broken down to release energy stored as ATP — the molecule cells use to power all their activities, including muscle contraction. During moderate exercise, this happens through aerobic respiration, which requires oxygen: Glucose + Oxygen → Carbon dioxide + Water (+ ATP energy released) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (+ ~38 ATP) Aerobic respiration is efficient and produces no harmful waste products. However, during the intense step-ups, your muscles were demanding energy faster than your respiratory and circulatory systems could deliver oxygen. When oxygen supply runs short, your muscle cells switch to anaerobic respiration — respiration without oxygen: Glucose → Lactic acid (+ a small amount of ATP energy) Anaerobic respiration produces far less ATP than aerobic respiration and releases lactic acid as a by-product. Lactic acid accumulates in the muscle tissue and lowers the pH, which is what causes that burning, aching sensation in your legs after the step-ups. The amount of oxygen your body needs after exercise — to break down the accumulated lactic acid and restore normal conditions — is called the oxygen debt. This is why you kept breathing heavily for several minutes even after you stopped exercising: your body was 'repaying' the oxygen debt by using that extra oxygen to convert lactic acid back into glucose in the liver.

Section 5: Answering the Driving Question — Putting It All Together	
Now bring everything together. Write a complete, connected answer to the driving question: 'Why does my body work so much harder to get oxygen into my cells during exercise — and what happens inside those cells	When you began the step-ups, your muscle cells immediately increased their rate of aerobic respiration — breaking down glucose from food to produce ATP for muscle contraction. This rapidly consumed oxygen and produced carbon dioxide, disturbing the balance that existed at rest. Your body responded by working much harder to restore that balance. Your diaphragm and intercostal muscles contracted more forcefully and more frequently, increasing both tidal volume and ventilation rate — which is why your breathing rate doubled from about 14 to 28 breaths per minute. At the alveoli, the steep concentration gradient was maintained, allowing rapid diffusion of oxygen into the blood and carbon dioxide out of it. At the same time, your heart rate rose from 72 to over 120 bpm, pumping oxygenated blood to your muscles far more quickly. The respiratory and circulatory systems were working in

when there is not enough?' Your answer should trace the full story from the moment exercise begins to the burning sensation in your muscles and the recovery period afterwards. Reference your personal data from the step-up activity and at least TWO scientific processes by name.	concert as a single oxygen-delivery machine. However, during peak effort, your muscles demanded ATP faster than aerobic respiration could supply it — even with doubled breathing and heart rates. Your muscle cells switched to anaerobic respiration, producing lactic acid. This lactic acid caused the burning sensation in your legs. After you stopped exercising, you continued breathing deeply — repaying your oxygen debt — until the lactic acid was cleared and normal conditions were restored. In summary: your body works harder during exercise because your muscle cells need more ATP, which requires more oxygen, which requires faster breathing and a faster heartbeat to deliver it. When oxygen still cannot keep up with demand, anaerobic respiration fills the gap — but at the cost of lactic acid build-up and the burning pain that you felt. Every part of that 2-minute step-up experience — the heavier breathing, the racing pulse, the burning legs, and the slow recovery — was your body's integrated response to a single challenge: keep the cells supplied with energy.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy — Gaseous Exchange	Accurately describes all key structural features of the alveoli (surface area, thin walls, moist lining, capillary network) and correctly explains diffusion across a concentration gradient. Uses precise vocabulary (haemoglobin, oxyhaemoglobin, partial pressure/concentration gradient) with no errors.	Correctly describes most structural features of the alveoli and explains diffusion with generally accurate vocabulary. Minor omissions (e.g., moist lining not mentioned) but no significant errors.	Shows basic understanding that gas exchange occurs in the lungs but descriptions are vague or incomplete. May confuse diffusion with breathing mechanics or omit key structural features. Vocabulary is limited or used imprecisely.
Scientific Accuracy — Respiration (Aerobic and Anaerobic)	Provides correct word equations for both aerobic and anaerobic respiration. Accurately explains the role of ATP, the conditions under which each process occurs, the products and their effects (including lactic acid and oxygen debt). Clearly distinguishes between the two processes.	Provides mostly correct equations and explains the main differences between aerobic and anaerobic respiration. May have a minor error in products or miss the concept of oxygen debt but demonstrates solid conceptual understanding.	Shows awareness that two types of respiration exist but equations are missing or incorrect. May confuse respiration with breathing, or incorrectly state the products. Explanation of lactic acid and oxygen debt is absent or inaccurate.
Explanation of the Phenomenon — Connecting Data to Science	Explicitly references personal or class step-up data (e.g., specific breathing rate and pulse rate values before and after exercise) and uses this data as evidence to support every section of the explanation. The link between observed changes and underlying biology is clear and logical throughout.	References the step-up activity and uses data in at least two sections. The connection between observed changes and biology is mostly clear but may lack specificity in one or two places.	Mentions the step-up activity but does not use specific data as evidence. Observed changes (faster breathing, higher pulse) are described but not convincingly connected to biological mechanisms.
Integration — Linking the	Clearly and accurately explains how the	Explains that the two systems are	Treats the respiratory and circulatory

Respiratory and Circulatory Systems	respiratory and circulatory systems work together during exercise. Explains why both breathing rate and heart rate rose simultaneously, what the blood transports, and why the rate of transport matters. The explanation is connected and coherent.	connected and that both rates rise during exercise. The reason for the connection (oxygen delivery and CO_2 removal) is present but may lack full detail about the feedback mechanism or the role of carbon dioxide.	systems as separate topics. May state that both rates rose but does not explain the biological reason for the connection. The explanation reads as a list of facts rather than an integrated account.
Communication — Use of Scientific Language and Structure	Consistently uses correct scientific vocabulary throughout (at least 8 key terms used accurately). Writing is well-organised, flows logically from phenomenon to mechanism to conclusion, and is clearly written for a scientific audience. The answer to the driving question is explicit and complete.	Uses correct scientific vocabulary in most sections (5–7 key terms). Writing is organised and mostly logical. The driving question is answered but the conclusion could be more explicit or complete.	Uses some scientific vocabulary but with frequent errors or vague usage (fewer than 5 key terms, or terms used incorrectly). Writing lacks clear organisation; the driving question is only partially answered or the answer is implied rather than stated.